A counterexample to signed cycle-restriction positivity

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Abstract

The signed cycle-restriction positivity conjecture fails at $n = 8$, $k = 3$, including the corrected formulation in the published paper. The coefficient of $x_1x_4x_6x_7y_2y_3y_5y_8$ is -1. We prove this by an exhaustive six-row table derived directly from the required permutation statistics. A second coefficient is $+1$, so changing a global sign cannot restore positivity.

Task 1. Full solution

The intended published statement

For a permutation $\pi \in \mathfrak{S}_n$, written as $\pi(1)\cdots\pi(n)$, define

$$\mathrm{RLMv}(\pi) = \{\pi(j) : \pi(j) < \pi(l) \text{ for every } l > j\}, \qquad \mathrm{EXCv}(\pi) = \{\pi(j) : \pi(j) > j\}.$$

For a set S, put $x_S = \prod_{s\in S} x_s$, and similarly for y_S. Let $c(\pi)$ denote the number of cycles. Write

$$P_{n,k}(\mathbf{x},\mathbf{y}) = (-1)^{n-1} \sum_{\substack{\pi\in\mathfrak{S}_n \\ \text{every cycle of } \pi \text{ has length} \geq k}} (-1)^{\mathrm{inv}(\pi)} x_{\mathrm{RLMv}(\pi)} y_{\mathrm{EXCv}(\pi)}. \tag{1}$$

Conjecture 5.2.1 of the journal version [1, p. 411] asserts $P_{n,k} \in \mathbb{N}[\mathbf{x},\mathbf{y}]$ for $2 \leq k \leq n$. The current preprint's Conjecture 40 and the MathDB transcription omit $(-1)^{n-1}$ and allow $k = 1$. The journal corrects both points. Our counterexample uses the journal's corrected statement.

Theorem 1. *For $n = 8$, $k = 3$, let*

$$M = x_1x_4x_6x_7y_2y_3y_5y_8, \qquad N = x_1y_2y_3y_4y_5y_6y_7y_8.$$

Then $[M]P_{8,3} = -1$ and $[N]P_{8,3} = 1$. In particular, the published conjecture is false, and the unnormalized polynomial in MathDB also has coefficients of both signs.

Deriving the complete table

To contribute to $[M]P_{8,3}$, a permutation must have

$$\mathrm{RLMv}(\pi) = \{1,4,6,7\}, \qquad \mathrm{EXCv}(\pi) = \{2,3,5,8\}. \tag{2}$$

It must in particular have no fixed points. Let $p_v = \pi^{-1}(v)$. The excedance conditions for the values $2, 3$ force $p_2 = 1$, $p_3 = 2$. The rightmost entry is the largest right-to-left minimum value, so $\pi(8) = 7$. Value 6 is not an excedance and is not fixed, hence $p_6 > 6$. Since position 8 is occupied,

$p_6 = 7$. Similarly $p_4 > 4$. The right-to-left minima occur in increasing value order, so $p_4 < p_6 = 7$; therefore $p_4 \in \{5, 6\}$. Finally 5 is an excedance value, giving $p_5 < 5$, and positions $1, 2$ are occupied; thus $p_5 \in \{3, 4\}$.

We have reduced every possible permutation to

$$\pi = 2\,3\,\square\,\square\,\square\,\square\,6\,7,$$

where the middle four entries are $1, 4, 5, 8$, subject to the conditions just stated and $p_1 < p_4$. If $p_4 = 5$, the values $1, 5$ fill positions $3, 4$ in either order, and 8 fills position 6. If $p_4 = 6$, place 5 at position 3 or 4, and fill the remaining two positions with $1, 8$ in either order. These give exactly the following six rows. Direct inspection confirms that each has the two statistic sets in (2).

One-line form	Disjoint cycles	$c(\pi)$	Contribution to $[M]P_{8,3}$
23154867	(123)(45)(687)	3	excluded: a 2-cycle
23158467	(123)(45876)	2	-1
23514867	(12354)(687)	2	-1
23518467	(12358764)	1	$+1$
23581467	(1235)(4876)	2	-1
23851467	(12387645)	1	$+1$

Indeed $\operatorname{sgn}(\pi) = (-1)^{8-c(\pi)}$, so the weight including the global factor in (1) is $(-1)^{c(\pi)-1}$. All cycles in the last five rows have length at least 3. Their contributions add to $-1 - 1 + 1 - 1 + 1 = -1$. This proves the first coefficient assertion without relying on computer enumeration.

The other sign and the source correction

For the monomial N, all values $2, 3, \ldots, 8$ must be excedances. Induction forces $p_v = v - 1$ for $2 \le v \le 8$: the earlier positions are already occupied by the smaller excedance values. Thus the unique permutation is $23456781 = (12345678)$. Its right-to-left minimum set is $\{1\}$ and its normalized sign is $+1$. Therefore $[N]P_{8,3} = 1$, completing the proof. □

For the unnormalized polynomial printed in MathDB, the coefficients of M, N are $+1, -1$, respectively. Hence this resolves the intended research assertion even after the journal's corrections; the missing global sign is not the mathematical obstruction established here.

Verification and attribution

The conjecture, statistics, and the known $k = 2$ result are due to Alexandersson and Getachew Kebede [1, 2]. Our six-row coefficient calculation disproves their proposed extension. The accompanying exact verifier checks the table and inversion signs, enumerates all 8! permutations, independently generates the allowed permutations by their cycles, and checks the known derangement identity and all smaller parameters. These computations supplement the complete table argument above.

All 23 pages of the current primary, all 27 pages of the journal final, and all nine pages of the proceedings version were read. The later refinements by Pei and Zeng [3] concern the derangement case; their stated results do not settle the cycle-length restriction. Bounded public checks found no earlier full claim for this target. Review is by the authoring agent, with no independent peer review or proof-assistant certification.

Tasks 2–5. Proposed follow-up directions

Task 2. Construct infinite families of negative coefficients for fixed $k = 3$, extending the six-row configuration above.

Task 3. Classify the pairs (n, k) for which all coefficients of $P_{n,k}$ are nonnegative; the cases $k = 2$ and $k > n/2$ are known or immediate.

Task 4. Determine which identifications of the x and y variables restore positivity, and whether the fully bivariate specialization has nonnegative coefficients for every cycle restriction.

Task 5. Describe precisely how forbidding 2-cycles removes sign-canceling partners in the source involutions, and derive a formula for the resulting negative terms.

References

[1] P. Alexandersson and F. G. Kebede, *An involution on derangements preserving excedances and right-to-left minima*, Australas. J. Combin. **86**(3) (2023), 387–413. Journal PDF.

[2] P. Alexandersson and F. G. Kebede, same title, arXiv:2105.08455v3, 2022.

[3] Y. Pei and J. Zeng, *Counting derangements with signed right-to-left minima and excedances*, arXiv:2206.11236v2, 2023.